

Dr. JEREMIAS IVAN

Canberra, ACT, Australia | jeremias.ivan98@outlook.com | <https://jeremiasivan.github.io>

PROFILE

Postdoctoral researcher in Evolutionary Biology, specialising in phylogenomic methods development to improve gene tree inference from genomic data, with a strong research and publication record spanning phylogenomics, cancer genomics, and computational biology. Seeking a research fellowship in bioinformatics, phylogenomics, or genome-wide association studies.

EDUCATION

Doctor of Philosophy in Evolutionary Biology

Aug 2022 - Jul 2026

Lanfear Group, Australian National University (ANU), Canberra, Australia

- Thesis: *New Methods to Improve Phylogenomic Inference*
- Supervisors: Prof. Robert Lanfear, A/Prof. Minh Bui, Prof. Justin Borevitz

Bachelor of Science in Bioinformatics

Sep 2016 - Oct 2020

Indonesia International Institute for Life Sciences (I3L), Jakarta, Indonesia

- GPA: 3.92 / 4.00 (*cum laude*)
- Thesis: *Comparative Phylogeography and Molecular Evolution of Australian Eugongylinae Skinks*
- Supervisors: Rizky Nurdiansyah, Prof. Craig Moritz, Dr. Sally Potter, Dr. Xia Hua

RELEVANT SKILLS

- **Bioinformatics:** phylogenomic inference for whole-genome data (e.g., IQ-TREE, ASTRAL); phylogenomic method development to detect systematic errors (e.g., contamination, reference bias); assessment of gene tree topological variation (e.g., ILS, introgression)
- **Programming and tools:** R (R Markdown, Quarto), UNIX, Git, high-performance computing (HPC)
- **Languages:** Indonesian (native), English (fluent)

RESEARCH AND TEACHING EXPERIENCE

Technical Assistant, Borevitz Lab [Casual]

May 2026 - Present

ANU, Canberra, Australia

- Reconstructed the phylogeny of eucalypts from whole-genome short-read data of more than 1,000 samples representing 500 described species
- Developed taxonomic consistency test to identify mislabeled and/or contaminated samples based on gene tree topological variation

Research Assistant, Moritz Lab [Casual]

Jun - Jul 2026

ANU, Canberra, Australia

- Reconstructed the phylogeny of sexual diploid and asexual triploid samples of *Heteronotia* geckos based on exon-capture data
- Identified gene conversion events and recombination events in the genomes of asexual triploid samples

Final Exam Markers, Gene Expression and Regulation (BIOL2161) [Casual]**Jun 2026***ANU, Canberra, Australia*

- Marked written final examination sheets of undergraduate students following the course rubrics, in coordination with the course convenor (Prof. Marcin Adamski) and other lecturers

Tutor, Data Science for Biologists (BIOL3207/6207) [Casual]**2023 - 2026***ANU, Canberra, Australia*

- Assisted undergraduate and graduate students for three semesters with data science in Biology using R and GitHub, including weekly readings and research idea development for the final project
- Marked and provided feedback on students' submitted works
- Discussed regularly with the course convenor (Prof. Robert Lanfear) and other tutors

Demonstrator, R + Quarto Workshop [Casual]**Apr 2024***ANU, Canberra, Australia*

- Assisted undergraduate and graduate students with data science in Biology using R, including data analysis using *tidyverse* package and data presentation using *Quarto*

Cancer Epigenetics Pipeline Development**Oct 2020 - Apr 2021***I3L, Jakarta, Indonesia*

- Developed automated epigenetics pipeline to download, filter, and identify differentially-expressed cancer biomarkers from The Cancer Genome Atlas (TCGA) dataset
- Supervisor: A/Prof. David Agustriawan

Credit Internship and Thesis Writing**Jul 2019 - Oct 2020***ANU, Canberra, Australia*

- Assessed how climatic variation over time and space influences the evolution and diversity of Australian Eugongylinae skinks
- Supervisors: Prof. Craig Moritz, Dr. Rust Turakulov, Dr. Sally Potter, Dr. Xia Hua

Non-credit Internship**Jun - Aug 2018***Asia University, Taichung, Taiwan*

- Developed automated pipeline to detect differentially-expressed methylation probes in cervical cancer patients based on TCGA dataset
- Supervisors: Prof. Ka-Lok Ng, Dr. Ezra B. Wijaya, Dr. Effendi Zainuddin, Dr. Eskezia Yihunie

OTHER PROFESSIONAL EXPERIENCES**Back End Developer****Feb - Dec 2022***Wardaya College, Jakarta, Indonesia*

- Led REST API development using Golang (Gin framework, GORM) and managed apps deployment in AWS Elastic Beanstalk
- Collaborated with senior developer on database migration, including ERD reconstruction, AWS RDS instance setup, and data migration

- Developed mobile and web applications using Golang (backend) and AWS/PostgreSQL, with regular discussion with CEO, apps designer, and Marketing Team

PUBLICATIONS

Preprints

- **J. Ivan**, B.S. Martin, K. Murray, A.-C. Colin, C. Kuelheim, R. Andrew, D. Nicolle, Z.A. Myburg, J. Leebens-Mack, A. Jones, R. Lanfear, & J.O. Borevitz. (2026). *Comprehensive Phylogenomic Inference of Eucalypts Reveals Taxonomic Limits through Gene Tree Concordance*. bioRxiv.
- **J. Ivan** & R. Lanfear. (2026). *PhyloRBT: A Phylogenetic Approach to Detect Reference Bias in Phylogenomic Datasets*. bioRxiv.
- **J. Ivan** & R. Lanfear. (2026). *PhyloNOW: Using Variable Window Sizes for Phylogenomic Analyses of Whole Genome Alignments*. bioRxiv. [Presented at AES 2025]

Journal Articles

- **J. Ivan**, P. Frandsen, & R. Lanfear. (2026). *Selecting a Window Size for Phylogenomic Analyses of Whole Genome Alignments Using AIC*. Systematic Biology, 75(1), 100-114. [Presented at RSB HDR Conference, ANU]
- **J. Ivan**, C. Moritz, S. Potter, J. Bragg, R. Turakulov, & X. Hua. (2022). *Temperature predicts the rate of molecular evolution in Australian Eugongyline skinks*. Evolution, 76(2), 252-261.
- **J. Ivan**, G. Patricia, & D. Agustriawan. (2021). *In silico study of cancer stage-specific DNA methylation pattern in White breast cancer patients based on TCGA dataset*. Computational Biology and Chemistry, 92, 107498. [Presented at InCoB 2020]
- D. Agustriawan, A. Parikesit, R. Nurdiansyah, **J. Ivan**, & K. Ramanto. (2021). *Correlation and transcriptomic analysis revealing potential microRNA-gene interactions associated with breast cancer formation*. Research Journal of Biotechnology, 16(2), 16-23.
- **J. Ivan**, D. Agustriawan, A. Parikesit, & R. Nurdiansyah. (2021). *MiRNA-Regulated HspB8 as potent biomarkers in low-grade gliomas*. Research Journal of Biotechnology, 16(1), 17-25. [Presented at BioMIC 2019]
- **J. Ivan**, R. Nurdiansyah, & A. Parikesit. (2020). *Comprehensive Molecular Simulation of Triple-negative Breast Cancer Transcriptomics Features of miR-145 and the 3' UTR of ARF6 mRNA*. International Journal Bioautomation, 24(3), 245-254.
- R. Kemal, **J. Ivan**, E. Sandjaja, & A. Santosa. (2020). *Computational Design of Ancestral and Consensus Asian Dengue Envelope Protein for Vaccine Candidate*. KnE Life Sciences, 2020.
- **J. Ivan**, R. Nurdiansyah, & A. Parikesit. (2020). *Computational modeling of AGO-mediated molecular inhibition of ARF6 by miR-145*. Indonesian Journal of Biotechnology, 25(2), 102-108.
- **J. Ivan**, R. Nurdiansyah, & A. Parikesit. (2019). *Mathematical Problem Solving: One Way to Prevent Dementia*. Iranian Journal Of Medical Informatics, 8(1), e10.
- S. Wibowo, D. Agustriawan, **J. Ivan**, A. Parikesit, & R. Nurdiansyah. (2019). *Mining Potential MicroRNA Biomarkers related to IQGAPs of Thyroid Carcinoma through in silico process*. Journal of Physics: Conference Series, 1192(1), 012070.
- **J. Ivan** & A. Parikesit. (2019). *Computational software for assessing allelic dropout*. Indonesian Journal of Life Sciences, 1(1).

- **J. Ivan**, D. Agustriawan, S. Wibowo, A. Parikesit, & R. Nurdiansyah. (2019). *IQGAP and HspB8: potent biomarkers in low grade gliomas*. Journal of Physics: Conference Series, 1192(1), 012056. [Presented at ICoDIS 2018]
- R. Kemal, E. Sandjaja, A. Santosa, & **J. Ivan**. (2018). *Short Communication: Identification of Mildew Locus O (MLO) genes in Durio zibethinus genome corresponding with the Powdery Mildew disease*. Biodiversitas Journal of Biological Diversity, 19(6), 2204-2212.

AWARDS

- A\$150 Travel Grant, AES 2025 (Oct 2025)
- Hirata Naora Best Presentation Awards, RSB HDR Conference, ANU (Apr 2024)
- Australian Government Research Training Program (AGRTP) Stipend Scholarship and International Fee Offset Scholarship, ANU (Aug 2022)
- Best Graduate of Bioinformatics with GPA = 3.92, I3L (Oct 2020)
- ANU Future Research Talent Awards 2020 at Moritz Lab, ANU (Mar 2020)
- Best Paper, Bioinformatics and Data Mining Symposium, BioMIC 2019 (Sep 2019)
- Partial scholarship (50% of total tuition fees), I3L (Aug 2016)